

Missouri S&T
MISSOURI UNIVERSITY OF SCIENCE AND TECHNOLOGY
Founded 1870 | Rolla, Missouri

Section 1.9

The Matrix of a Linear Transformation

Identity Matrices

2×2 identity matrix: $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
 Its columns are $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

3×3 identity matrix: $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
 Its columns are $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, and $\mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

Identity Matrices

$n \times n$ identity matrix: $I_n = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}$
 Its columns are $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$.

When the size is clear from the context, we often refer to the identity matrix as simply I .

Theorem

If $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation, then there exists a unique matrix A such that

$$T(\mathbf{x}) = A\mathbf{x}$$

for all $\mathbf{x}$ in $\mathbb{R}^n$.

Specifically, A is the $m \times n$ matrix

$$A = [T(\mathbf{e}_1) \cdots T(\mathbf{e}_n)]$$

where $\mathbf{e}_j$ is the j th column of the $n \times n$ identity.

A is called the standard matrix for the linear transformation T

Example 1

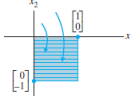
Prove the existence portion of the preceding theorem.

Example 2

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation that rotates each point in $\mathbb{R}^2$ about the origin through an angle φ , where positive angles correspond to counterclockwise rotation. Find the standard matrix A of this transformation.

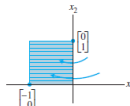
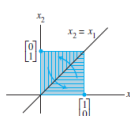
Common Transformations

TABLE 1 Reflections

Transformation	Image of the Unit Square	Standard Matrix
Reflection through the x_1 -axis		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

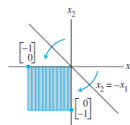
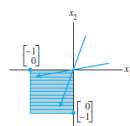
Common Transformations

TABLE 1 Reflections

Transformation	Image of the Unit Square	Standard Matrix
Reflection through the x_2 -axis		$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$
Reflection through the line $x_2 = x_1$		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

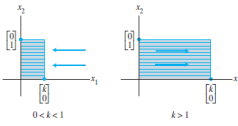
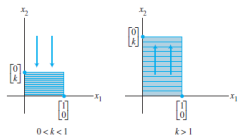
Common Transformations

TABLE 1 Reflections

Transformation	Image of the Unit Square	Standard Matrix
Reflection through the line $x_2 = -x_1$		$\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$
Reflection through the origin		$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

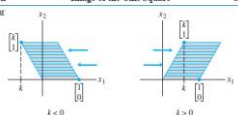
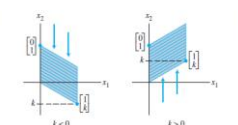
Common Transformations

TABLE 2 Contractions and Expansions

Transformation	Image of the Unit Square	Standard Matrix
Horizontal contraction and expansion		$\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$
Vertical contraction and expansion		$\begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$

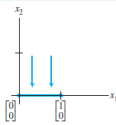
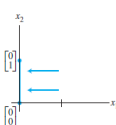
Common Transformations

TABLE 3 Shears

Transformation	Image of the Unit Square	Standard Matrix
Horizontal shear		$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$
Vertical shear		$\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$

Common Transformations

TABLE 4 Projections

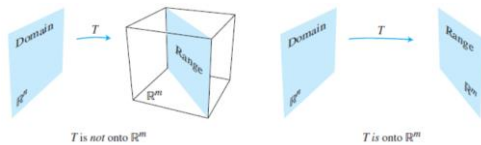
Transformation	Image of the Unit Square	Standard Matrix
Projection onto the x_1 -axis		$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$
Projection onto the x_2 -axis		$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

One-To-One and Onto Transformations

A mapping $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be onto $\mathbb{R}^m$ if each $\mathbf{b}$ in $\mathbb{R}^m$ is the image of **at least one** $\mathbf{x}$ in $\mathbb{R}^n$.

Equivalently, T is onto if its range is equal to its entire codomain.

One-To-One and Onto Transformations



One-To-One and Onto Transformations

A mapping $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be one-to-one if each $\mathbf{b}$ in $\mathbb{R}^m$ is the image of **at most one** $\mathbf{x}$ in $\mathbb{R}^n$.

Equivalently, T is one-to-one if, for each $\mathbf{b}$ in $\mathbb{R}^m$, the equation $T(\mathbf{x}) = \mathbf{b}$ has either a unique solution or none at all.

Example 3

Let T be the linear transformation whose standard matrix is

$$A = \begin{bmatrix} 1 & -4 & 8 & 1 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

Does T map $\mathbb{R}^4$ onto $\mathbb{R}^3$?

Is T a one-to-one mapping?

Theorem

Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then, T is one-to-one if and only if the equation $T(\mathbf{x}) = \mathbf{0}$ has only the trivial solution.

Theorem

Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation, and let A be the standard matrix for T . Then,

- T maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ if and only if the columns of A span $\mathbb{R}^m$ (which occurs if and only if A has a pivot in every row); and
- T is one-to-one if and only if the columns of A are linearly independent (which occurs if and only if A has a pivot in every column).

Example 4

Show that T is a linear transformation by finding a matrix that implements the mapping. Then, determine if the transformation is one-to-one and/or onto.

$$T(x_1, x_2) = (3x_1 - 2x_2, x_1 + 4x_2, x_2)$$
